Doc. 263 – Fractal Holography: FFGFT and the Holographic Principle

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Contents

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Preliminary remark

This document locates FFGFT relative to the holographic principle. It strictly separates three levels: **(1)** contact points already documented in the FFGFT corpus; **(2)** an open scaling question at the area law, computed here as a *reduction layer* but *not* claimed as a core derivation; and **(3)** genuine tensions where FFGFT explicitly does *not* match the standard picture of holography. The separation is deliberate: otherwise the word "holographic" loads an AdS/CFT expectation that FFGFT neither meets nor intends to meet.

Methodological status. Section .1 reports what is documented (Doc. 201). Section .2 computes a consequence of *one assumption* and is marked as a reduction layer. The choice of assumption is a physical decision, not a derivation.

Symbols used.

Symbol	Meaning
ξ	basic geometric parameter, $\xi = 4/30000 \approx 1.333 \times 10^{-4}$
$\tilde{T} \cdot m = 1$	time–mass duality (fundamental relation); $\tilde{T}$ intrinsic time scale
D_f	fractal dimension, $D_f = 3 - \xi$
E_0	characteristic energy, $E_0 = \sqrt{m_e m_\mu} = 7.398$ MeV
E_H	Hubble energy, $E_H = E_0\, \xi^{41/4} \approx 1.41 \times 10^{-33}$ eV
H_0	Hubble quantity, $H_0 = E_H/\hbar \approx 66.2$ km/s/Mpc
R_H	maximal scale, $R_H = \hbar c/E_H \approx 1.4 \times 10^{26}$ m
L_0	sub-Planck granulation, $L_0 = \xi \ell_P \approx 2.2 \times 10^{-39}$ m
L_ξ	characteristic ξ length ($\sim 100\,\mu$m, Doc. 091)
ℓ_P	Planck length
z, z_{obs}	redshift; $z_{\text{obs}} = \lambda_b/\lambda_e - 1$ (wavelength ratio)
$\rho_{\text{vac}}, \rho_{\text{CMB}}$	vacuum/CMB energy density; $\rho_{\text{CMB}} = \xi\hbar c/L_\xi^4$
m_T, λ	mediator mass $m_T = \lambda/\xi$ (Doc. 201); regulates $\rho \le 1/\xi^2$
$c, \hbar, \hbar c$	SI conversion factors (not ontological primitives)
T_{00}	energy density from $\mathscr{L} = \xi(\partial E)^2$; $T_{00} = \xi[(\partial_t E)^2 + (\nabla E)^2]$

Layers: core derivation > bridge > reduction layer (plausibility). Standing caution: H_0, z, Ω are ΛCDM pipeline outputs, not model-neutral.

.1 Documented contact points

Area law from node counting (Doc. 201, §7.6)

FFGFT sets the entropy of a black hole as node configuration entropy:

$$S = \frac{A}{4\,\ell_P^2} \;\longrightarrow\; S = N_{\text{nodes}} \ln 2, \qquad N_{\text{nodes}} \propto \frac{A}{\xi_0^2},$$

with $\ell_P^2 \sim \xi_0^2$ in the large limit. This reproduces the Bekenstein-Hawking area law – the holographic core statement that entropy scales with *area*, not *volume*. In FFGFT this is not a postulate; it falls out of counting saturated nodes on the core surface.

Information as an access problem, not an existence problem

The winding numbers $(n_\theta, n_\varphi, n_3, n_4)$ are topological invariants; they cannot disappear. An apparent "information loss" at the horizon is, in FFGFT, epistemic (a problem of access), not ontological (a problem of existence). This is a distinct position on the information paradox, entering the same terrain as holography – conservation of information – but by a different mechanism: topological conservation rather than boundary encoding.

Mass as scale projection

FFGFT already uses the term "holographic projection": particle masses arise as a projection between scales (the VEV onto the sub-Planck scale). This is holography in the sense of a *projection between scales*, not in the sense of a bulk/boundary duality.

.2 Open scaling question at the area law (reduction layer)

Decompactification forces corrections – but not their form

A natural argument runs: since we live in a decompactified regime (the torus dimensions are hidden at the scale $L_0 \sim \ell_P/\xi_0$), the hidden compactness *must* show up as a fractal correction in observable quantities. This is correct – and it is precisely ξ_0: the correction parameter is documented in the corpus (mass formula $m_i = r_i\, \xi_0^{p_i}\, v$; geometric redshift as fractal path-lengthening). That fractal corrections exist *at all* is therefore not speculation.

This argument does *not*, however, fix *which* geometric quantity the correction acts on at a black-hole horizon. "Decompactification forces corrections" does *not* entail $S \propto R^{2-\xi_0}$. A correction on the horizon *area* (Assumption B below) and one on the enclosed *volume* (Assumption C) are both fully decompactified and fractal – yet they yield *different* area laws (broken vs. exact). The step from "ξ_0 corrections exist" to "the horizon entropy deviates by $\sim 1\,\%$" is therefore an *additional* assumption about where the correction acts, not a consequence of decompactification itself.

The question

Doc. 201 uses the *plain* area law: $N_{\text{nodes}} \propto A \propto R^2$, exponent exactly 2. A correction by the fractal dimension $D_f = 3 - \xi_0$ is *not* built in there. It is therefore an open question whether D_f affects the horizon *area* at all – and if so, how.

Three assumptions compared

- **Assumption A (plain area law, Doc. 201):** $S \sim R^2$. Exponent exactly 2.
- **Assumption B (fractal area):** if the nodes sit on a codimension-1 boundary of a D_f-dimensional structure, the effective node count scales as $R^{D_f-1} = R^{2-\xi_0}$. Exponent $2-\xi_0$.
- **Assumption C (fractal volume):** if D_f acts on the enclosed *volume* ($V \propto R^{D_f}$) while the area stays integer, the area law remains *exact* at exponent 2.

Result of the computation

Assumption	Exponent	Deviation from area law
A – plain area law	2	0
B – fractal area	$2-\xi_0 = 1.99986667$	$\Delta_{\text{exp}} = \xi_0 = 1.33 \times 10^{-4}$
C – fractal volume	2	0 (area law stays exact)

Only Assumption B breaks the Bekenstein-Hawking law – by *exactly* ξ_0 *in the exponent*. The relative deviation grows logarithmically with size:

$$\frac{S_B}{S_A} - 1 \;\approx\; -\xi_0 \,\ln\!\left(\frac{R}{\ell_P}\right).$$

The tiny ξ_0 is lifted by the logarithm across many orders of magnitude; for astrophysical black holes the effect reaches percent level:

Object	R/ℓ_P	Deviation (Assumption B)
Solar-mass BH ($R \sim 3$ km)	1.9×10^{38}	$-1.17\,\%$
Sagittarius A* ($R \sim 1.2 \times 10^{10}$ m)	7.4×10^{44}	$-1.37\,\%$
M87* ($R \sim 1.9 \times 10^{13}$ m)	1.2×10^{48}	$-1.47\,\%$

Interpretation

This is qualitatively different from the ξ_0 effect in other contexts (e.g. the dimension-driven proximity $D/(D+1) \to 3/4$, where ξ_0 only appears at 10^{-5}). Here ξ_0 reaches percent level because $\ln(R/\ell_P)$ bridges some 38 orders of magnitude. This is precisely where a *discriminator* would sit between "true holography" (area law exact) and "fractal near-holography" (exponent $2-\xi_0$).

Status (open). The choice between Assumption B and C is a physical decision about FFGFT, *not* a derivation from the existing corpus. Assumption B is plausible (codimension-1 boundary), but it is not derived in Doc. 201. As long as the question is open, the percent deviation is the consequence of *one* assumption, not a prediction of FFGFT. Likewise, $\ell_P^2 \sim \xi_0^2$ was treated here as a pure prefactor (it cancels in the ratio); any scale-dependent fractality of ℓ_P would add a further term.

.3 Reconstruction: the hologram analogy and its limits

Documented: a non-invertible projection (Doc. 193)

Doc. 193 records that evaluating the mode labels (r_i, p_i) to real numbers is *not invertible*: once r_i and p_i have been evaluated to numbers in $\mathbb{R}$, they carry "no memory" of their origin from different particles. From a real value (a mass, say, or a derived quantity) the underlying quantum-number structure cannot be uniquely recovered – the fibre over each numerical value has collapsed. This is a documented, non-invertible projection from the discrete geometric mode space onto $\mathbb{R}$.

Analogy (metaphor, not derivation): the physical hologram

This asymmetry can be illustrated by a physical hologram – explicitly as an *illustration*, not a derivation. An optical hologram encodes a spatial scene in an interference pattern on a surface. To turn it back into a visible image, the surface alone does not suffice: it requires a reference (readout) beam carrying the *prior information* (wavelength, phase, recording geometry). And even at the best recording and reconstruction quality, the visible hologram remains a *faint image* of the original reality – part of the information is in principle unrecoverable.

On this reading, the observable $3{+}1$-dimensional world relates to the underlying T^4 geometry as a reconstructed hologram relates to the recorded scene: reconstruction requires prior information, and it is lossy in principle. The observable world is an *incomplete* image of the substrate – not because the recording is poor, but because the projection itself discards information.

Distinction: this is not the non-closure theorem

This non-invertible projection must be carefully separated from the *non-closure theorem* (Doc. 193). The non-closure theorem is a *different*, purely algebraic statement: the set of admissible pairs (r_i, p_i) is not closed under multiplication, i.e. the product of two distinct modes $(r_i \cdot r_j,\ p_i + p_j)$ corresponds to no particle of the model. It concerns the *combinability* of modes with one another, not the *invertibility* of a projection. Both statements come from Doc. 193 but are distinct and should not be equated.

.4 Reversibility: an assumption of standard holography vs. a derived time asymmetry

Reversibility is an axiom in standard holography

Standard holography (AdS/CFT) is formulated as a *duality* – an exact, invertible dictionary between bulk and boundary. This invertibility rests on two premises: the **unitarity** of the underlying quantum theory (reversible time evolution, information conservation) and the existence of a **conformal boundary** on which the exact encoding lives. Both are posited, not derived. Reversibility is therefore not a measured finding but a *structural assumption* of the formalism.

FFGFT derives the time asymmetry (Doc. 157)

FFGFT does not treat the direction of time as an axiom but derives it from deeper geometry. Doc. 157 anchors the breaking of time-reversal symmetry in three ways:

- **Algebraic:** $T \cdot m = 1$ with positive-definite mass ($m > 0$, antimatter included) forces $T > 0$; a reversal $T \to -T$ would require $m \to -m$ and is physically impossible.
- **Topological:** the winding of the vacuum field on the torus has a chirality; once chosen, the time direction is topologically protected (like a knot that cannot be reversed by continuous deformation).

- **Fractal:** at $D_f = 3 - \xi$ the path integrals for forward- and backward-running paths differ – the fractal structure turns the discrete symmetry $t \to -t$ into a continuous asymmetry.

The two situations are therefore *not* symmetric: standard holography *posits* reversibility, whereas FFGFT *derives* non-reversibility. Non-invertibility is thus not a separate extra assumption in FFGFT but a consequence of structures the framework already carries ($T \cdot m = 1$, torus winding, D_f).

Empirical limit: only practical experience

Honesty is required here. The only empirical anchor for the directedness of time at all is the **practical experience** of directed happening – not a direct measurement and not a theorem. The observed CP/T violation is, as Doc. 157 itself notes, too weak for the macroscopic arrow of time. FFGFT thus does not *measure* the asymmetry; it replaces the thermodynamic stopgap (an unexplained low-entropy initial condition) with a *geometric derivation* of the directedness already experienced in practice. This is a structural advantage – the "Past Hypothesis" becomes unnecessary – but it is not an empirical proof of the premise.

Consequence: reversibility as a possible artefact

From this follows the genuine open question lying beneath both frameworks. *If* the world is fundamentally time-asymmetric – as practical experience suggests and FFGFT derives geometrically – *then* the exact reversibility of the holographic mathematics is an **artefact of the unitary idealization**: it abstracts away a symmetry breaking that FFGFT grounds. This is not a claim that standard holography is "wrong", but the observation that its reversibility is an unmeasured assumption – and that FFGFT posits a derived asymmetry at the same point. The two assumptions have different observable consequences; a concrete discriminator has already been named (the logarithmic entropy deviation $-\xi \ln(R/\ell_P)$ under Assumption B, Section .2).

.5 Anchoring at the bounded core (no singularity)

The anchor

A projection is admissible as a pure model; but for it to be *FFGFT-intrinsic* (rather than borrowed from AdS/CFT), it must be *anchored* at a physical point, like the time-mass duality. T0 has no singularity: the vacuum density is bounded by $\rho \leq 1/\xi^2 \approx 5.6 \times 10^7$ (Doc. 201), regulated by the mediator mass $m_T = \lambda/\xi$, which prevents unbounded growth of Δm. Where classical GR would run into a singularity, the ξ-bounded core holds against it – exactly as $T \cdot m = 1$ caps the divergence ($m \to \infty$ is capped by $T \to 0$). The holographic projection is anchored at this core.

Computational consequence: a cutoff instead of a singularity

This entails a *physical cutoff*. There is a finite core radius R_{core} below which the nodes are saturated; the area-law scaling $S \sim R^a$ holds only for $R \geq R_{\text{core}}$, and below it the entropy saturates. Order of magnitude:

$$\frac{R_{\text{core}}}{\ell_P} \sim \frac{1}{\xi} \approx 7.5 \times 10^3 \qquad (\log_{10} \approx 3.9).$$

The $O(1)$ prefactor is model-dependent; only the *order of magnitude* follows from $\rho \leq 1/\xi^2$. Consequence for the computation in Section .2: values with $\log_{10}(R/\ell_P) \lesssim 3.9$ lie *below* the core – in the region of the non-existent singularity – and are physically meaningless. Only above R_{core} is the scaling valid. For astrophysical black holes ($R/\ell_P \sim 10^{38}$–$10^{48} \gg R_{\text{core}}$) the deviation $-\xi \ln(R/\ell_P)$ remains valid unchanged – the cutoff affects only the unphysical small-scale region.

Interpretive limit: conserved information is geometric base information

What is conserved at the core – the node configuration entropy $N_{\text{nodes}} \ln 2$ – is *not* a maximal description of the past reality, but only the **base information about the geometric structure** (winding numbers as topological invariants). Conservation here means "access to the geometry", not "reconstruction of the history". This is consistent with the access-not-existence character (Section .2) and with the lossy, directed projection: the history of the infalling material is not conserved countably, only its geometric signature.

Open: no exact translation quantity

A gap remains to be stated. So far we have only a *scale anchoring* ($R_{\text{core}} \sim \ell_P/\xi$), not a derived **translation quantity** that would gauge the area law (core entropy ↔ horizon area) as an exact map with a fixed coefficient – analogous, say, to the central charge in AdS/CFT or to the SI conversion factors $\hbar, c$ within FFGFT itself. No such quantity is documented in the corpus, and none has been found. This is consistent with the position taken here: no exact duality, hence no exact translation coefficient – only a lossy projection with a scale anchoring. The exact translation quantity remains an open point.

Boundaryless topology

Standard holography (AdS/CFT) requires a conformal *boundary*. The universal 4-torus T^4 is boundaryless (compact without boundary). A naive bulk/boundary duality therefore does not apply. FFGFT's "holography" is an area law plus scale projection – not boundary encoding.

No holographic dark energy

Models that obtain Λ from a holographic horizon are incompatible with the static FFGFT universe (no Λ, no dark energy). Here FFGFT stands in *opposition* to holography, not in contact. An equation of state w_{DE} does not exist in FFGFT; the comparable observable is the geometric redshift law (fractal path-lengthening, cf. Doc. 041/182), not a w parameter.

.6 H_0 as a projection artefact, not a translation quantity

No direct ξ–H_0 bridge – but a chain through $\hbar c$

The opposition to holographic dark energy tempts one to ask whether FFGFT's ξ and the ΛCDM parameter H_0 can be *translated* into one another. A *direct* identification $\xi/L_0 = H_0/c$ fails – but not because energy and length were incompatible. In natural units ($\hbar = c = 1$) length and energy are dual; the passage to SI metres is exactly one conversion factor ($\hbar c = 1.973 \times 10^{-7}$ eV m), and in FFGFT $\hbar, c$ are SI conversion factors anyway, not ontological primitives. The error of the naive bridge is not the energy–length duality but the *choice* *of scale*: it couples ξ to the sub-Planck granulation $L_0 = \xi \ell_P \approx 2.2 \times 10^{-39}$ m instead of to the cosmological scale $R_H \approx 1.4 \times 10^{26}$ m. Both lengths convert correctly into energies via $\hbar c$ – they are simply *different* scales (ratio $R_H/L_0 \approx 6.5 \times 10^{64}$, which is calibrated to the measured H_0, not following from ξ alone).

The correct translation chain: $\hbar c$ as the SI factor

Through $\hbar c$ the chain does close (Doc. 182):

$$\xi \;\longrightarrow\; E_H = E_0\,\xi^{41/4} \;\xrightarrow{\times\hbar c}\; R_H = \frac{\hbar c}{E_H} \;\longrightarrow\; H_0 = \frac{c}{R_H} = \frac{E_H}{\hbar}.$$

With $E_0 = \sqrt{m_e m_\mu} = 7.398$ MeV this gives $E_H = 1.412 \times 10^{-33}$ eV, $R_H = 1.398 \times 10^{26}$ m and $H_0^{T0} \approx 66.2$ km/s/Mpc (-1.9% from the Planck value). Here $\hbar c$ is precisely the SI conversion factor that turns the energy representation (E_H) into the length representation (R_H) – exactly the quantity one expects when length is readable as energy in natural units. So ξ fixes the energy *and* – via $\hbar c$ – the associated length; only the *cosmological* length R_H, not the sub-Planck length L_0.

The mechanism: H_0 is a misread slope

This chain is at the same time the *explanation* of H_0 as an artefact – consistent with the projection reading of holography. The mechanism: for small z the fractal path-lengthening is linear, $z(d) = \kappa d$ with $\kappa = E_H/(\hbar c)$. An observer in the expansion picture reads the same linear $z(d)$ curve and *names* its slope H_0/c. Hence

$$H_0^{\text{(measured)}} = \kappa c = E_H/\hbar,$$

i.e. H_0 is not a dynamical expansion parameter but the *geometric coefficient* of the path-lengthening, misinterpreted as expansion. The artefact is the *reinterpretation of a slope*, not a measurement error. The translation chain $\xi \to E_H \to R_H \to H_0$ does exist – it only shows that the translated quantity is geometric, not dynamical. Beyond the maximal scale R_H one has $z \to \infty$; no ΛCDM-type particle horizon is needed.

H_0 is emergent, not fundamental

What matters is where H_0 sits ontologically: it is *not a fundamental constant* of the theory. In the compactified T^4 form there is no H_0 – there are no cosmological distances there, no vacuum traversed as a medium, and $\hbar, \varepsilon_0$ become "real" only upon decompactification (Doc. 262). Since $H_0 = E_H/\hbar$ contains this only-then-real $\hbar$, H_0 is an *emergent* quantity of the decompactified description – just as age, critical density and Hubble length are *Friedmann concepts without a direct T0 counterpart* (Doc. 016). Only ξ is fundamental; H_0 arises at the transition compact → decompactified. The non-integer exponent $41/4$ expresses precisely this scale transition (scale flow over ~ 40 orders of magnitude), not a pure geometric constant; forcing an integer reading would wrongly push an emergent transition quantity back into the compact geometry. (Earlier formulations calling H_0 a "fundamental constant" or the redshift an "energy loss in the vacuum" stem from an outdated stage; cf. Doc. 190, R16.)

The same holds for z itself: real view, misreading, translation

The reinterpretation concerns not only the slope H_0 but the whole z value – and here the clean three-way split is worthwhile.

Model-neutral (real): what is measured is a wavelength ratio, $1 + z_{\text{obs}} = \lambda_{\text{obs}}/\lambda_{\text{emit}}$. This quantity is observable and identical in both pictures; it is not an artefact.

The real FFGFT view: z_{obs} is the accumulated *fractal path-lengthening* of the photon over the physical distance d, $z_{\text{obs}} = (L_{\text{eff}} - d)/d$ (Doc. 026). No scale factor, no expansion, no time evolution of space – a purely geometric path-length relation in the static T^4.

The misreading (ΛCDM): the same z_{obs} is read as an *expansion* redshift, i.e. as a cosmological scale factor $1 + z = a_0/a_{\text{emit}}$. In this reading the initial linear slope becomes H_0 and a large value such as $z \approx 1100$ becomes the "decoupling" epoch of a recombination. Both are the expansion reading of the *same* geometric path-lengthening – not a second, independent measurement (cf. P15/R15, Doc. 190; the static theory has no cosmological z in the expansion sense).

Translatable: the two views convert into one another because they describe the *same* $z_{\text{obs}}(d)$ curve. H_0 is the slope at small z, $z \approx 1100$ an extreme value near the maximal scale R_H; both are ΛCDM readings of the one FFGFT geometry (the FFGFT-own value of this extremum is $z_* \approx 875$, Doc. 267; $z \approx 1089/1100$ is only the expansion reading, not the FFGFT value). The translation is therefore a *reinterpretation of the curve*, not a physical identity of parameters: what is real is z_{obs}, what is an artefact is solely its expansion reading.

Placement and open point

This is precisely the resolution of the apparent dark-energy opposition: a translation *does* exist – the chain $\xi \to E_H \to R_H = \hbar c/E_H \to H_0$ – but no *direct* $\xi \leftrightarrow H_0$ identity at the sub-Planck scale (which would obscure its pipeline character and contradict the standing caution). The model-neutral shared quantity remains the observable $z(d)$: same curve, two mechanisms, distinguishable at large z. **Open – and to be placed:** the exponent $41/4$ is not freely chosen but measures the *scale transition* from the characteristic energy E_0 to the Hubble energy E_H via the scale-flow/recursion structure of the ξ field (Doc. 026/182, recursion R Doc. 203). Two things must be separated: (i) the *structure* of the chain $\xi \to E_H \to R_H \to H_0$ is documented and consistent – H_0 is therein a geometric coefficient misread as expansion, not a dynamical parameter; $\hbar$ acts only as SI conversion ($H_0 = E_H/\hbar$), not as a physical input (cf. section "Constants as projection factors"). (ii) the *numerical value* of the exponent $41/4$ itself is not yet fully derived from the ξ-field theory; Doc. 165 discusses a unit-free integer form $n = 10$ (with several possible geometric readings, itself open). As long as (ii) is open, the H_0 *value* is a reduction layer – a strong consistency argument (-1.9% from the Planck value), not a closed core result. The *interpretation* (H_0 as a projection artefact) does not depend on (ii) and stands independently.

.7 Constants as projection factors – and the vacuum energy

The H_0 reading is a special case of a more general principle that recurs across several quantities and supports the holographic projection idea quantitatively. It runs in four steps; all four are reproducible as short calculations and are flagged below as reduction layer.

1. The "constants" ground nothing – they convert

In natural units ($\hbar = c = 1$, pure ratios) the FFGFT description is static and correction-free (Layer 1, Doc. 261). The quantities c, $\hbar c$ and $\sqrt{\hbar G/c^5}$ are not physical inputs but *conversion factors* of the SI projection (Layer 2):

- c is the metre/second factor that arises when the quantities length and time – identical in Layer 1 ($\tilde{T} \cdot m = 1$) – are projected into separate units (Doc. 134: $c = L/T$, "not a law of nature but a ratio"). The value $299\,792\,458$ is the metre definition, not a ξ prediction.
- $\sqrt{\hbar G/c^5}$ grounds no time scale; it expresses the already-fixed scale in seconds (pure unit calibration).
- Even G follows c-free from ξ: $G_{\text{nat}} = \xi^2/(4m_{\text{char}})$ (Doc. 012/013); $c, \hbar$ enter only at the SI conversion.

2. The vacuum structure is fully present when compactified

The mode structure (ξ/L_ξ^4, fractal mode counting at $d = 3 + \delta$, Doc. 091) is *fully present* in Layer 1 – merely unit-free. *Nothing is added* at decompactification: the same structure appears in Layer 2, multiplied by the conversion factor $\hbar c$, as a dimensional energy density $\rho_{\text{vac}} = \xi \hbar c/L_\xi^4$ (form from Doc. 091/025). The vacuum energy thus does not *come into being* – it merely becomes *visible* from the SI view through the decompactification of the previously enrolled fractal/temporal structure.

3. No static energy floor: the energy is dynamical

That there is no *self-standing* (static) vacuum energy follows computationally from the energy-field Lagrangian $\mathscr{L} = \xi(\partial_\mu E)(\partial^\mu E)$ with field equation $\Box E = 0$ (massless, no potential term $V(E)$, Doc. 010). The canonical energy-momentum tensor gives

$$T_{00} = \xi\left[(\partial_t E)^2 + (\nabla E)^2\right].$$

T_{00} contains *no* static term. Switching off the temporal component ($\partial_t E = 0$) and requiring a source-free vacuum solution ($\Box E = 0 \Rightarrow \nabla^2 E = 0 \Rightarrow E =$ const) gives $T_{00} = 0$. The vacuum energy is therefore entirely bound to the temporal (and, under boundary conditions such as the Casimir effect, spatial) *variation* of the field – not an eigenvalue of a field at rest but an expression of the dynamics. The *free* vacuum has no spatial part; the *confined* one (Casimir) does, consistent with the free-vs-confined distinction in Doc. 091.

4. Delimitation: what of this is robust

Robust are the quantities in which ξ stands *alone* (no fit factors, no circular exponents): $T_{\text{CMB}} = \frac{16}{9}\xi^2 E_\xi$ and $|\rho_{\text{Casimir}}|/\rho_{\text{CMB}} = \pi^2/(240\xi) \approx 308$.

Clarification on the nature of this T_{CMB} relation. It is an *algebraic* ξ relation, *not a fractal correction*: the factor $16/9$ is simply $(4/3)^2$, the square of the numerical prefactor of $\xi = \frac{4}{3} \times 10^{-4}$ (cf. Doc. 025/028/006); $D_f = 3 - \xi$ and a mode count with fractal dimension do *not* enter this derivation. What is robust is the *ratio* $T_{\text{CMB}}/E_\xi = (16/9)\xi^2$ (Layer 1, unit-free). The absolute Kelvin value 2.725 K follows only via the SI conversion (embedding price, Doc. 261); the residual deviation of $\approx 0.9\%$ (Doc. 013) is *not* derived from ξ. A correction of the form $1 - C\xi$ that would close these 0.9% exists in the corpus only as a *numerical approximation* (e.g. $C = 275/4$), not as a derivation from the recursion $\mathscr{R}$ (Doc. 203) – it remains open.

The ratio given in Doc. 028, $\rho_{\text{DE}}/\rho_{\text{DM}} = \xi^{\ln(2.5)/\ln\xi}$, requires separate placement. It yields exactly 2.5 for *any* ξ and is therefore circular *as a prediction*. Yet its target value 2.5 ($\Omega_\Lambda/\Omega_{\text{DM}}$)

is itself not a model-neutral measurement but a ΛCDM-pipeline output. Read *not* as a prediction but as a *conversion factor* between the FFGFT geometry and the ΛCDM Ω bookkeeping – analogous to c (metre/second) and to z – the circular calibration via 2.5 is admissible. But only *partially*: unlike c, where both sides share the same real counterpart, the ΛCDM side carries with ρ_{DE} a model artefact (no Λ in FFGFT); the factor translates partly something real (DM bookkeeping $\leftrightarrow$ ξ effect), partly an artefact. Cf. Doc. 190, R17.

The *interpretation* is thereby clear: dark energy is eliminated (no Λ, static); dark matter is *not a substance* but a ξ-geometric effect – an artefact of the ΛCDM view, like H_0 and z. *Quantitatively* open is only the model-neutral reproduction of what ΛCDM books as DM (Ω_{DM}, rotation curves).

Status. This section is reduction layer: it makes a principle documented in Doc. 261/262 (Layer 1 correction-free / Layer 2 SI projection) explicit for c, ρ_{vac} and the dynamics. It predicts no new cosmological value; the numerical-value circularity of L_ξ (fixed from ρ_{CMB}) remains a limit.

.8 A formal bridge to UIFT – verifiable on one side only

The same projection picture yields a *formal* bridge to the information quantity ξ_{UIFT} of Onur Teker's UIFT (Doc. 257). It is placed here, not asserted: its force is *asymmetric*.

The bridge rests solely on the *first order* $T_{\mathrm{CMB}}/E_\xi = (16/9)\xi$ (not on the fractal correction K_{frak} and not on the absolute Kelvin value). Inserting it into the UIFT relation $\xi_{\mathrm{UIFT}} = k_B T_{\mathrm{CMB}} \ln 2/(m_e c^2)$ gives a *fixed* factor

$$\frac{\xi_{\mathrm{FFGFT}}}{\xi_{\mathrm{UIFT}}} = \frac{m_e c^2/\mathrm{eV}}{(16/9)\ln 2} \approx 414\,684.$$

This value is not an empirical result but a pure *conversion factor* between a Layer-1 quantity (ξ_{FFGFT}, dimensionless) and a Layer-2 quantity tied to the CMB temperature (ξ_{UIFT}); it coincides with its closed form.

What this verifies – and what it does not. The *FFGFT side* is fully reproducible: ξ is geometric, $(16/9)\xi$ follows from the mode count, m_e from the lepton hierarchy – anyone can recompute the factor. The *UIFT side* we *cannot* verify: the relation $\xi_{\mathrm{UIFT}} = k_B T_{\mathrm{CMB}} \ln 2/(m_e c^2)$ comes from Teker's work, and whether it is correct or empirically validated lies beyond our reach. The bridge is therefore *formal*: *if* Teker's relation holds *and* the FFGFT values hold, *then* a fixed factor links the two quantities. What is thereby verified is the FFGFT side of the bridge – not the UIFT side.

.9 Summary

- **Documented:** FFGFT reproduces the Bekenstein-Hawking area law from node counting (Doc. 201); information is topologically conserved (access, not existence problem); mass as scale projection.
- **Open (reduction layer):** if $D_f = 3-\xi_0$ affects the horizon area (Assumption B), the exponent becomes $2-\xi_0$ with a logarithmically growing deviation $-\xi_0 \ln(R/\ell_P)$ (percent level for real BHs). If D_f acts on the volume (Assumption C), the area law stays exact. The choice is to be made on physical grounds.
- **Reconstruction:** the projection of mode labels onto $\mathbb{R}$ is non-invertible (Doc. 193) – illustrated by the physical hologram (metaphor): reconstruction needs prior information and remains a faint image. Strictly separated from the non-closure theorem.

- **Anchoring:** the projection is anchored at the ξ-bounded core ($\rho \leq 1/\xi^2$, no singularity; Doc. 201) – like $T \cdot m = 1$. Consequence: a cutoff at $R_{\text{core}} \sim \ell_P/\xi$; below it the entropy saturates, the $R^{2-\xi}$ scaling holds only above. Real black holes lie far above (deviation valid). Conserved information = geometric base information, not the history. Open: no exact translation quantity, only a scale anchoring.
- **Reversibility:** standard holography *posits* invertibility (unitarity + boundary); FFGFT *derives* the time asymmetry (Doc. 157: $T \cdot m = 1$, torus winding, D_f). The only empirical anchor of directedness is practical experience; under that premise the exact reversibility is an artefact of the unitary idealization.
- **Tension:** T^4 is boundaryless (no AdS/CFT duality); no Λ, no w_{DE} (opposition to holographic dark energy).
- H_0 **artefact:** no *direct* $\xi \leftrightarrow H_0$ equation at the sub-Planck scale – the naive form fails on the choice of scale ($L_0 = \xi\ell_P$ instead of R_H), not on the energy–length duality (in natural units both are dual via $\hbar c$). But the chain $\xi \to E_H = E_0\xi^{41/4} \to R_H = \hbar c/E_H \to H_0 = E_H/\hbar$ does exist (≈ 66.2 km/s/Mpc; Doc. 026/182), with $\hbar c$ as the SI factor. H_0 is thus a geometric coefficient misread as expansion; the model-neutral quantity remains the observable $z(d)$. Open: exponent $41/4$ not yet fully derived.
- **Constants as projection:** c, $\hbar c$, $\sqrt{\hbar G/c^5}$ ground nothing; they convert to SI (Doc. 134/261). The vacuum structure is fully present when compactified (unit-free) and becomes SI-visible only on decompactification. $T_{00} = \xi[(\partial_t E)^2 + (\nabla E)^2]$ (from $\mathscr{L} = \xi(\partial E)^2$, $\Box E = 0$, Doc. 010) has NO static term – without field variation $T_{00} = 0$: no self-standing energy, the energy comes from the temporal component.

Terminological caution. "Holographic" in FFGFT is to be read as *area law + scale projection*, not as *boundary encoding*. This distinction should remain explicit in all FFGFT texts to avoid false AdS/CFT expectations.